SUPPLY CHAIN INSIGHT HUB DCT

1. OVERVIEW:

What is this use case about?
Answer:
The Supply Chain Insight Hub provides end-to-end visibility of the MA order-to-delivery process, including customer claims.
It enables tracking of performance against targets and identification of supply risks related to unallocated orders and free stock availability, supporting effective utilization for billing and target achievement.
The Order-to-Delivery cycle is broken down into individual stages, allowing detailed analysis and deeper insights at each step.

When is the report updated?
Answer:
Refresh - Three times a day (7:00, 13:00, 17:00)

KPI?
Answer: TNS Achievement, Order fulfillment, TAT Adherence

2. USER DOCUMENTATION:

Abbreviations (As per Dashboard)

CM Billing – Current Month Billing

BTD – Balance To Do

LT – Lead Time

MTS – Make To Stock

MTO – Make To Order

TO – Transfer Order

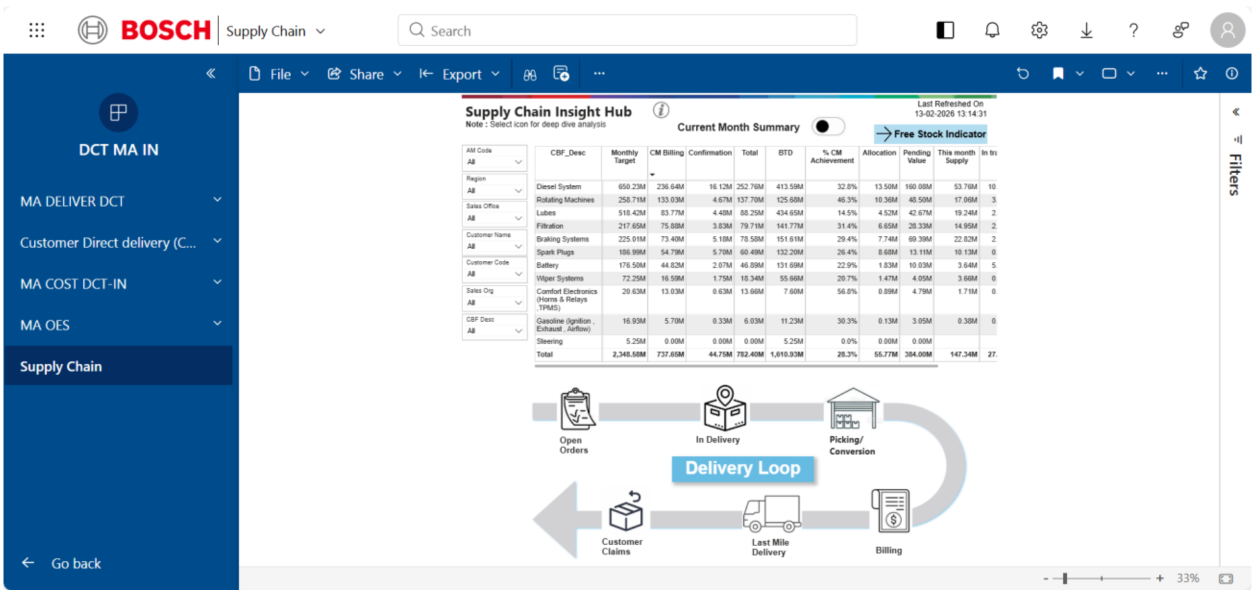
OBD – Outbound Delivery

MOM Billing – Month on Month Billing

EDD – Expected Delivery Date

TAT – Turnaround Time

This section describes the Supply Chain Insight Hub dashboard, which provides structured visibility of MA order-to-delivery execution, including performance against targets, delivery progress, billing status, claims, and free stock availability.



The Supply Chain Insight Hub provides operational visibility across order execution stages within the Delivery Loop and related supply chain monitoring blocks.

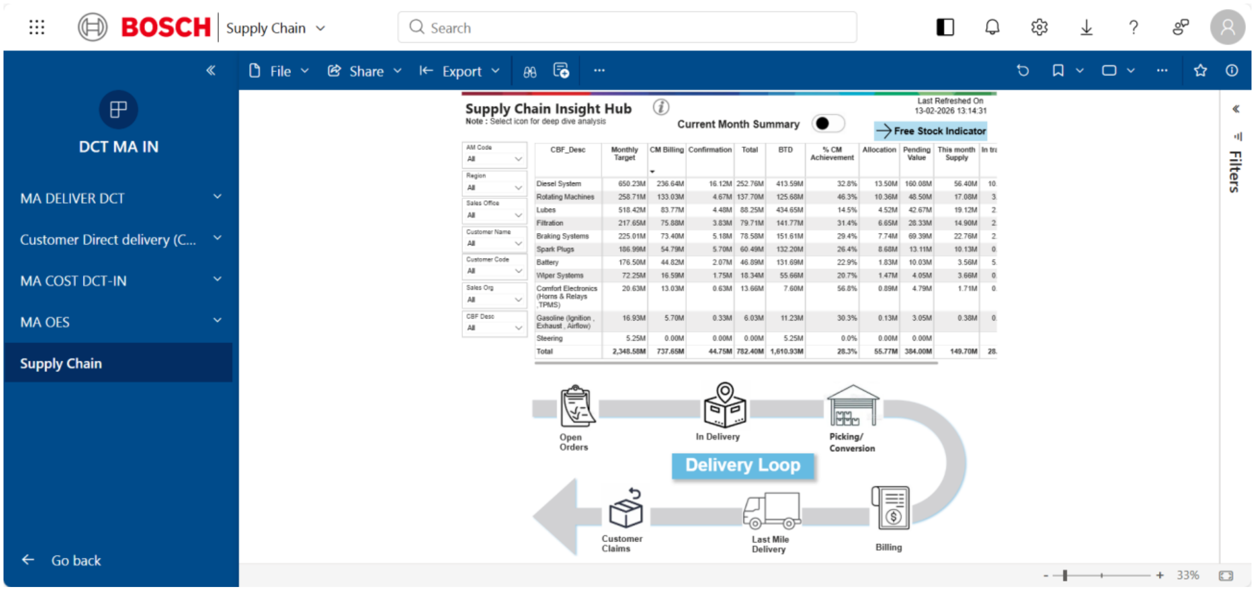
__**The dashboard is structured into:**__

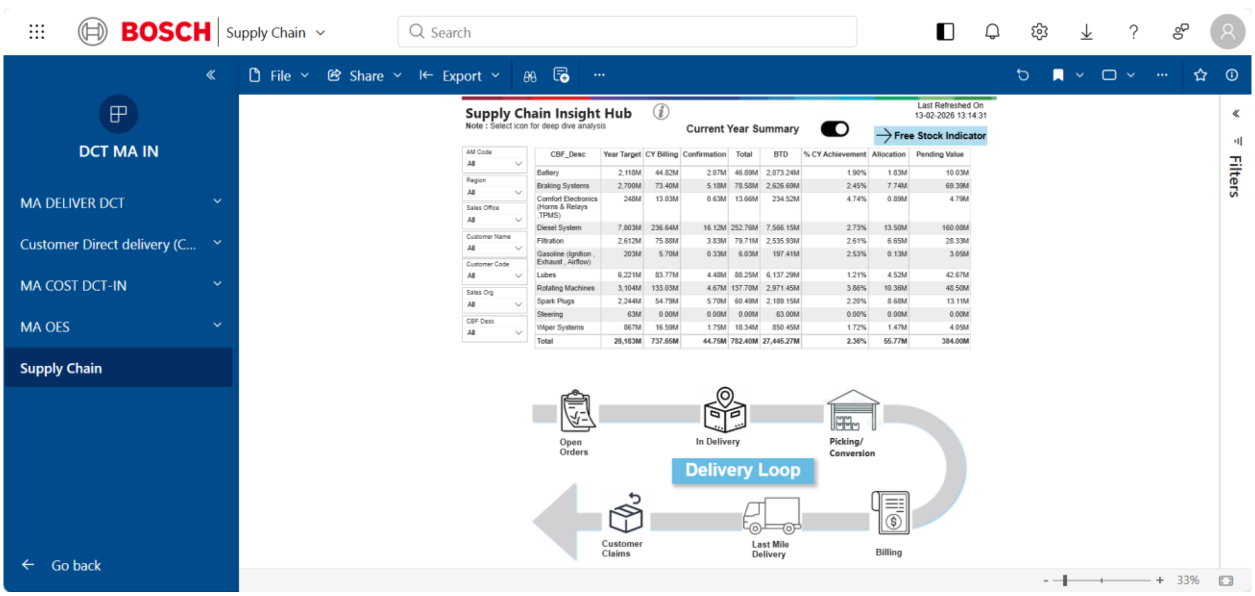
Current Month Summary/Current Year Summary

Delivery Loop Monitoring

Free Stock Indicator

1. Current Month Summary (Main Page)/Current Year Summary





Current Month Summary

Purpose:

To monitor current month performance against customer signed target at CBF level and assess supply risk from unallocated orders and stock gaps.

Why:

Enables MA OES teams to identify sales risk early, prioritize allocation, accelerate picking, and generate orders for parts with available free stock to protect target achievement.

Logic Explanation:

Target: Customer signed target from BRO portal

CM Billing: Billing as per IAM calendar (auto-adjusted to current month date)

Picking: Allocated orders with credit block released and pending warehouse conversion

BTD: MAX(Target - Billing - Picking, 0)

Allocation: Material available and delivery created, either under credit block or pending credit check

Plant-wise free stock visibility supports:

Sales team in identifying parts to generate orders

Order execution team in transferring orders to plants with available stock

Free Stock Logic:

If (Stock - Total Open Order) > 0 → Free Stock = YES (Green)
Else → Blank

Graph and Table Connection:

Main table provides CBF-wise Target, CM Billing, Confirmation, Total, BTD, Allocation, and Pending Value

% Achievement is derived from Target vs Billing

Allocation and Pending Value reflect execution backlog

Free Stock indicator highlights plant-wise availability to mitigate BTD risk

Current Year Summary

Purpose:

To monitor YTD progress against customer signed target at CBF level and identify balance to do.

Why:

Provides management visibility on cumulative performance trend and execution gap to achieve annual commitment.

Graph and Table Connection:

Same column logic as Current Month Summary.

YTD Billing replaces CM Billing.

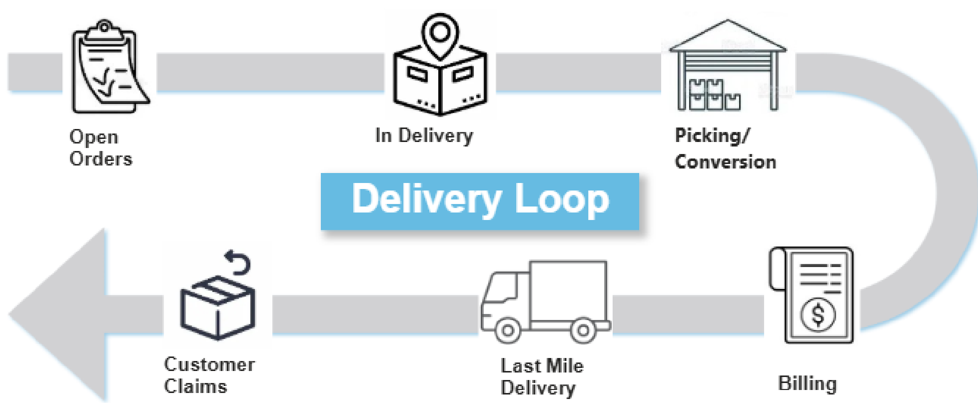
% Achievement represents YTD performance vs annual target.

BTD reflects remaining gap for the year.

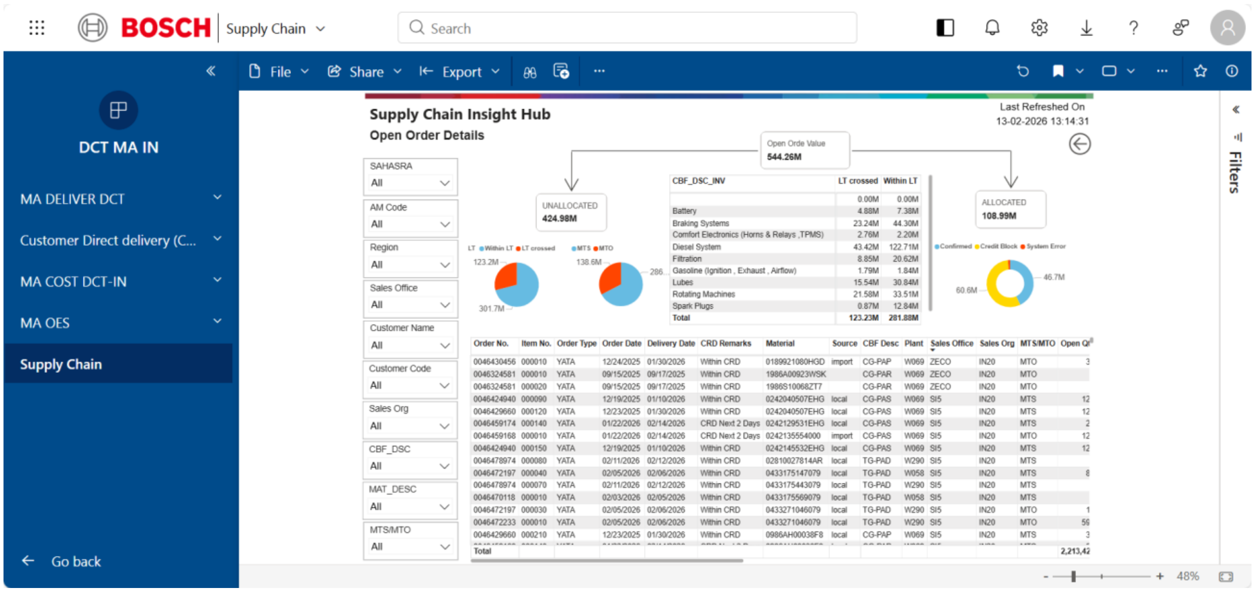
2. Delivery Loop

The Delivery Loop represents the operational flow from Open Orders to Customer Claims.

Open Orders → In Delivery (Allocation) → Picking/Conversion (Credit check completed and pending for warehouse to complete) → Billing → Last Mile Delivery (In transit & closed shipments) → Customer Claims



2.1 Open Orders



Purpose:

To monitor open sales orders and allocation status against Lead Time.

Why:

Helps identify LT deviations, unallocated orders, and confirmation gaps before they impact billing.

Graphs and Table Connection:

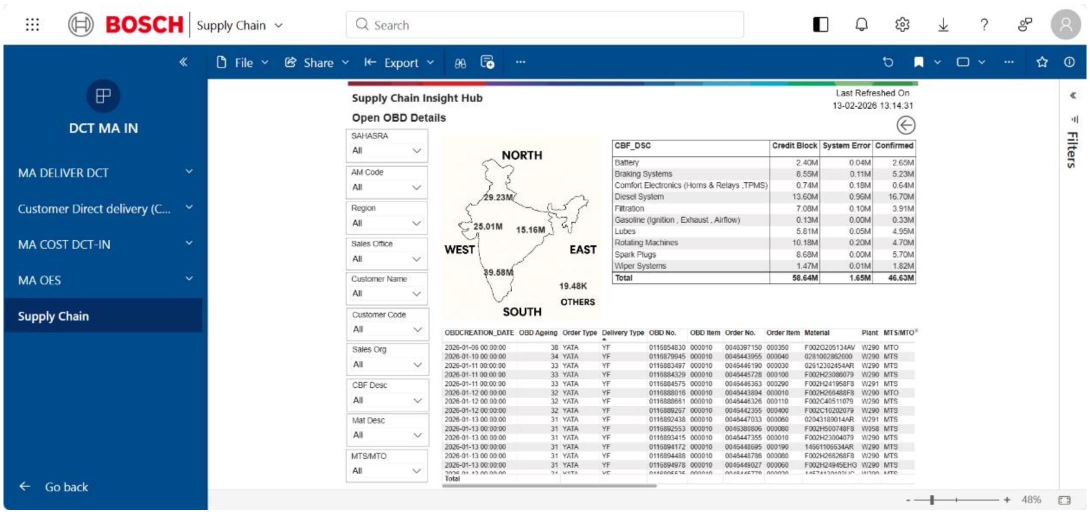
Pie charts represent LT crossed vs Within LT split.

MTS/MTO split shows supply model exposure.

CBF-wise distribution highlights high-risk product categories.

The detailed table lists order-level records that feed the summary visuals.

2.2 In Delivery



Purpose:

To monitor outbound deliveries, credit block/system error status, and regional execution distribution.

Why:

Ensures that confirmed deliveries are not stuck due to credit or system issues and supports region-wise execution control.

Graph and Table Connection:

Open OBD Value card reflects total open delivery value.

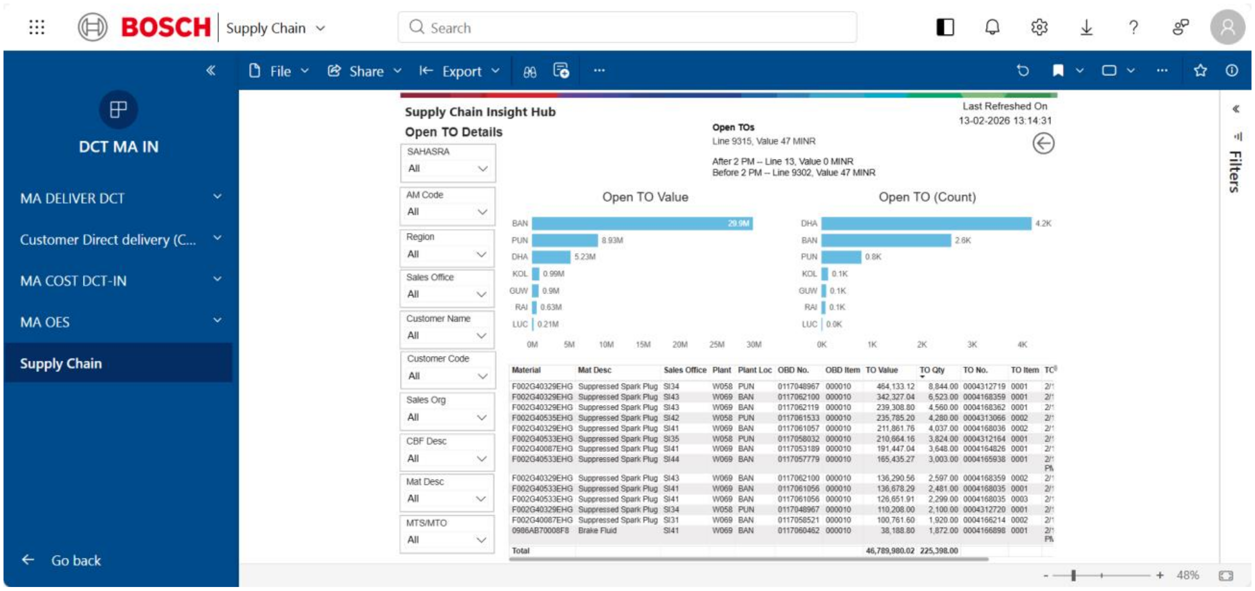
Donut chart shows Confirmed vs Credit Block vs System Error split.

Regional map displays execution distribution by geography.

CBF split highlights category exposure.

The OBD detail table contains delivery-level records feeding all visuals.

2.3 Picking / Conversion



Purpose:

To track Transfer Order creation and warehouse picking progress.

Why:

Identifies warehouse bottlenecks affecting conversion from delivery to billing.

Graph and Table Connection:

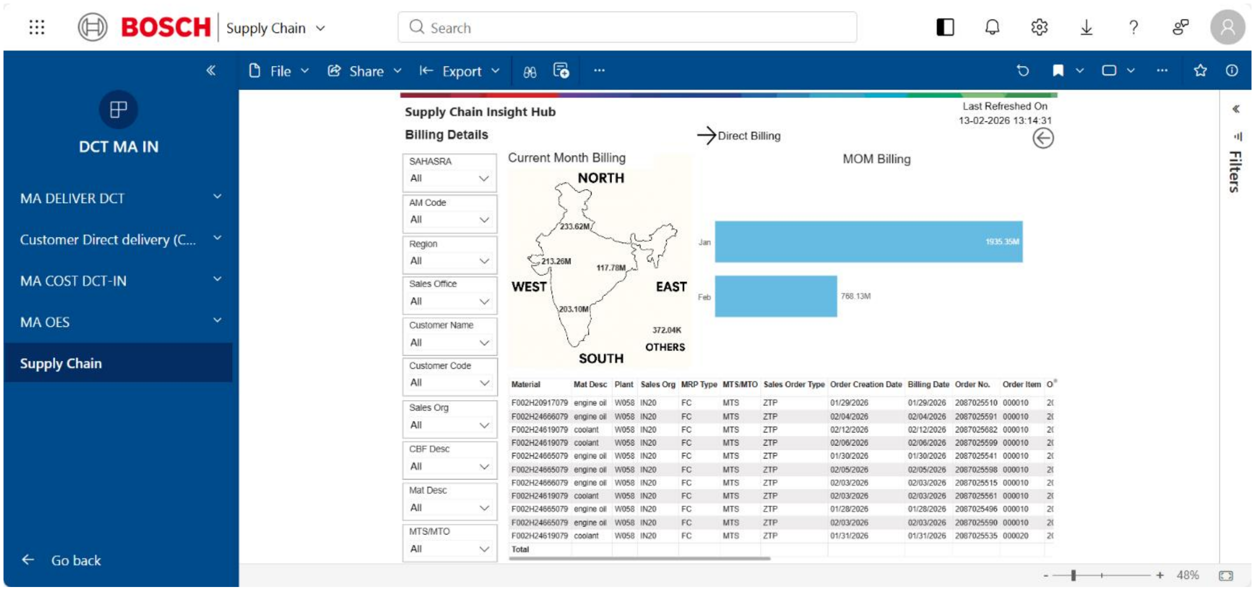
Open TO Value and Count represent total pending warehouse workload.

Before 2 PM / After 2 PM split indicates daily operational discipline.

Plant-wise distribution highlights load concentration.

Detailed TO table supports root cause analysis at material level.

2.4 Billing



Purpose:

To monitor current month billing execution and regional contribution.

Why:

Ensures billing acceleration in line with target and identifies region-wise performance gaps.

Graph and Table Connection:

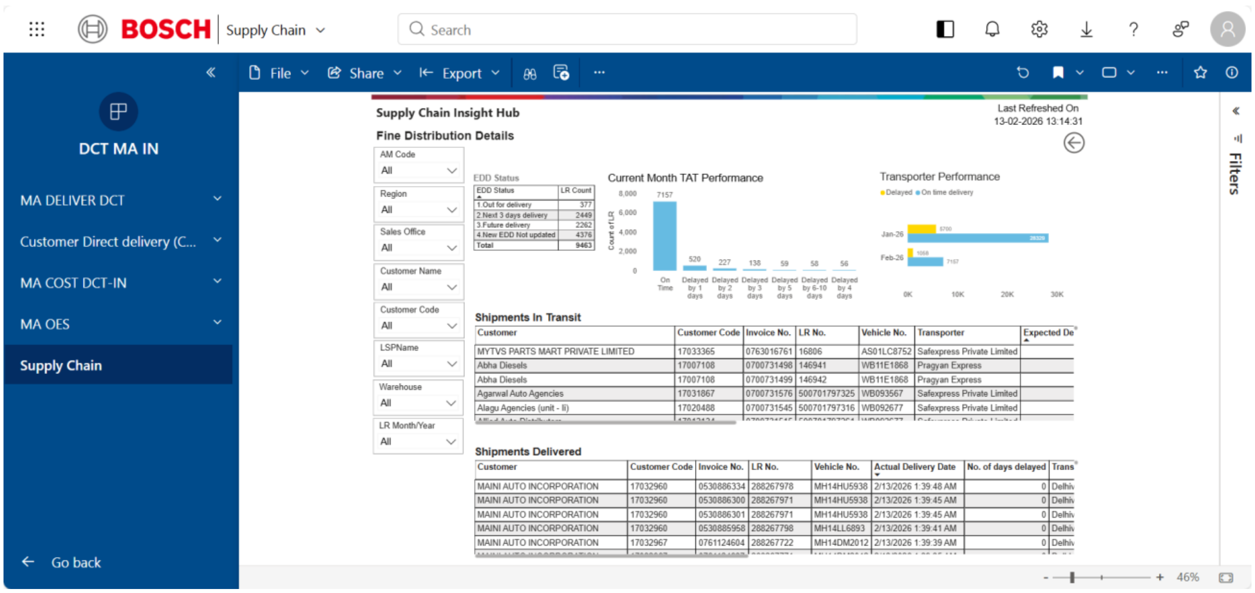
Region map shows billing distribution across North, West, East, South, Others.

MOM Billing chart compares current month vs previous month performance.

Direct Billing toggle isolates direct billing cases.

Billing detail table provides invoice-level records supporting the visuals.

2.5 Last Mile Delivery



Purpose:

To monitor delivery timeliness, transporter performance, and shipment status.

Why:

Ensures adherence to Expected Delivery Date and improves service level performance.

Graph and Table Connection:

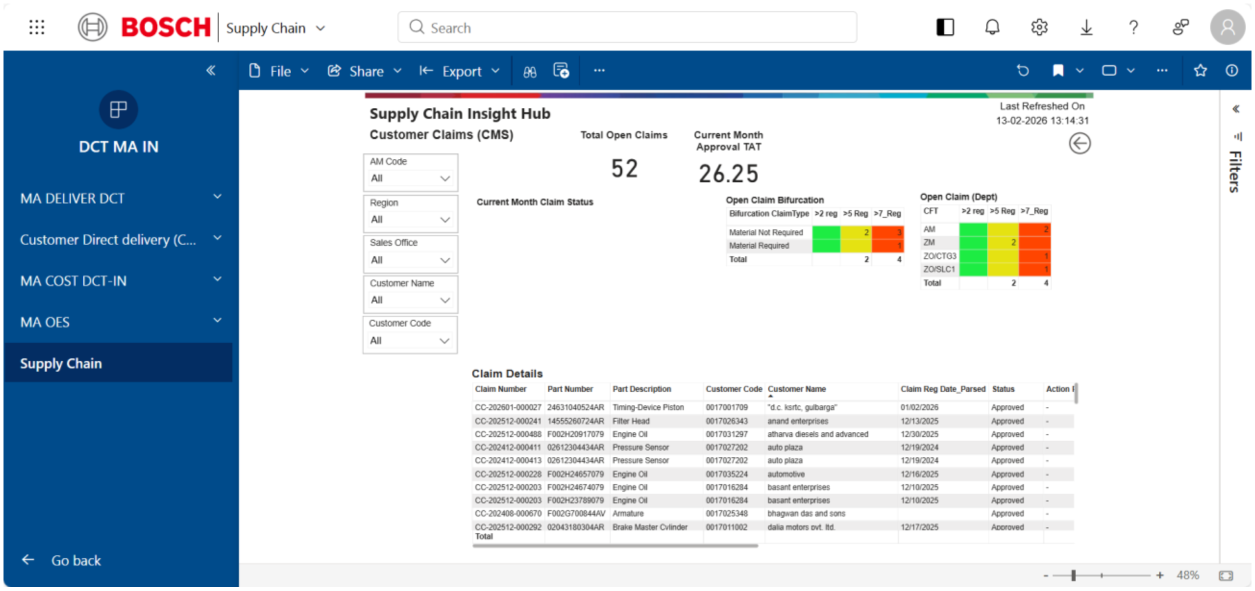
EDD Status chart shows On Time vs Delayed distribution.

TAT chart shows delay buckets.

Transporter performance chart compares On Time vs Delayed shipments.

Shipments in Transit and Delivered tables provide shipment-level visibility.

2.6 Customer Claims (CMS)



Purpose:

To monitor claim ageing, approval turnaround time, and departmental backlog.

Why:

Prevents claim ageing escalation and reduces revenue blockage.

Graph and Table Connection:

Total Open Claims card shows overall pending exposure.

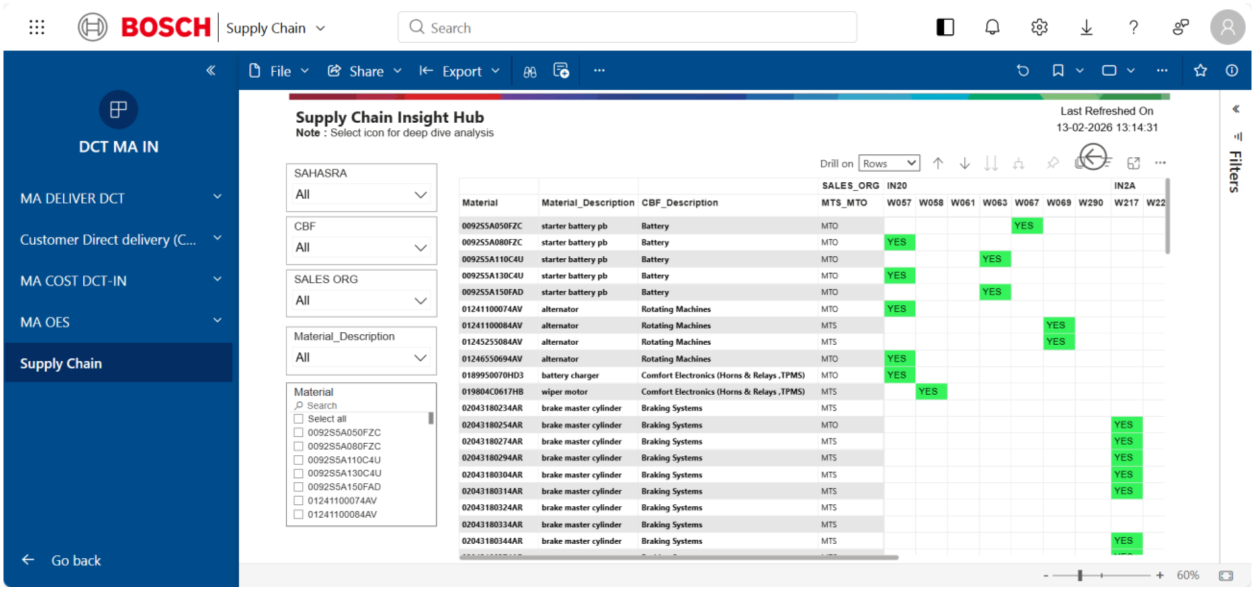
Current Month Approval TAT shows approval efficiency.

Claim bifurcation (>2 Reg / >5 Reg / >7 Reg) reflects ageing severity.

Department-wise claim heatmap identifies functional backlog.

Claim detail table provides claim-level transaction records feeding all summaries.

3. Free Stock Indicator



Purpose:

To provide plant-wise free stock visibility at part level.

Why:

Supports sales team in generating orders for available stock and enables order execution team to transfer orders to plants with stock availability.

Graph and Table Connection:

Matrix view shows Material, CBF, MTS/MTO classification and Sales Org.

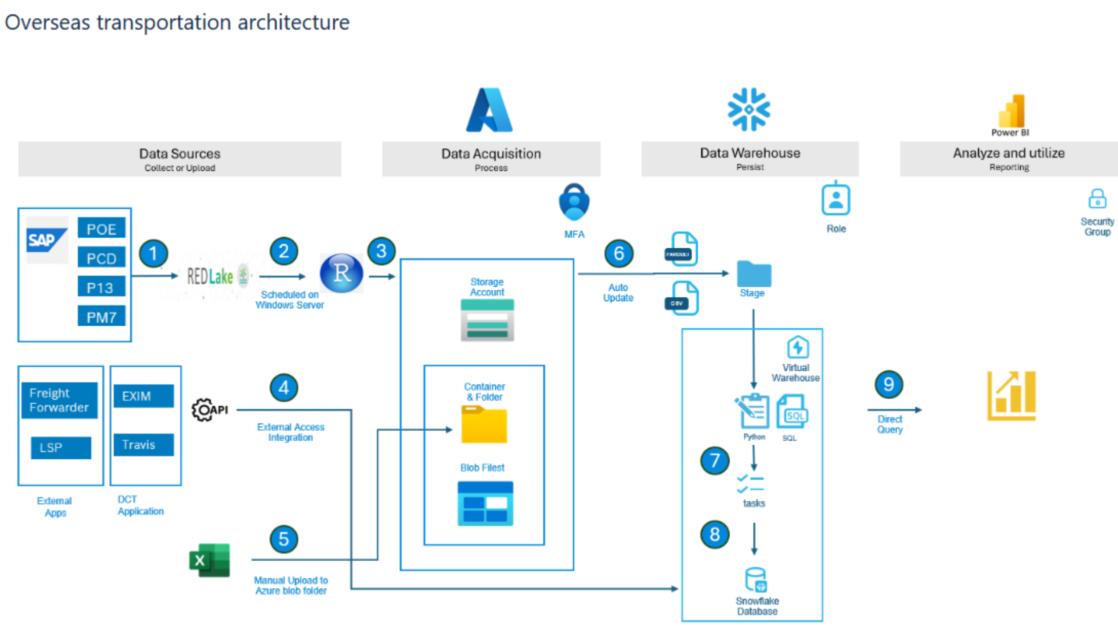
Plant-wise availability is displayed as YES indicator.

Logic: If (Stock – Total Open Order) > 0 → YES (Green), else blank.

Note: Free Stock Indicator is not part of the Delivery Loop.

3. <u>TECHNICAL DOCUMENTATION:</u>

This section briefly outlines the technical architecture underpinning the Bosch Supply Chain Insight Hub system, highlighting its data flow and key components.



3.1 Supply Chain Insight Hub Technical Architecture Overview

The Supply Chain Insight Hub system leverages a robust, cloud-enabled architecture to ensure reliable data acquisition, processing, storage, and reporting. It builds upon established Bosch data infrastructure to deliver actionable insights.

3.2 Key Architectural Phases

3.2.1 Data Sources:

SAP (POE, PCD, PL7): Core enterprise data from various SAP modules, covering procurement, commercial documents, and material management.

External Apps (Freight Forwarder, LSP, Travis, DCT App) & Excel: Integrates data from logistics partners, dedicated CDD applications, and allows for manual supplementary data uploads.

3.2.2 Data Acquisition:

RedLake & Windows Server: Extracts data from SAP via RedLake, with processes scheduled on Windows Servers for automation.

APIs & MFA: External application data is ingested via secure APIs. Multi-Factor Authentication (MFA) ensures secure access during data transfer.

Azure Storage Account: Data is stored in Azure Blob Storage (containers & folders) after initial acquisition, including manual uploads.

Auto Update: Automated processes ensure continuous data refresh within the storage account.

3.2.3 Data Warehouse:

Snowflake (Virtual Warehouse): Data is processed and persistently stored in Snowflake, a cloud data warehousing platform.

Staging: A temporary area for data validation and transformation.

Python & SQL: Used for complex data transformations, cleaning, and orchestration of data pipelines (tasks) within Snowflake.

3.2.4 Analyze & Utilize:

Power BI: The primary reporting tool, connecting directly to Snowflake via Direct Query. This enables near real-time dashboards and reports.

Security Groups: Access to Power BI reports is managed through security groups, ensuring data confidentiality and controlled access for authorized users.

This architecture ensures that the Supply Chain Insight Hub system is scalable, secure, and provides timely, accurate data for proactive decision-making.

-> Power BI Dashboard Link: Supply Chain - Power BI

-> Snowflake Default Role Setup: **Snowflake Default Role Setup - 3S IT - End User Guide - Docupedia**

-> How to get Access: **DCT Access.docx**

-> Apply for the required role: **oneIDM**

QUESTIONS + ANSWERS

1. What is the Supply Chain Insight Hub?

ANSWER:

The Supply Chain Insight Hub provides end-to-end visibility of the MA order-to-delivery process, including customer claims.

Business Purpose:

- Track performance against targets
- Identify supply risks
- Monitor unallocated orders
- Monitor free stock availability
- Support billing and target achievement

The dashboard breaks the Order-to-Delivery cycle into operational stages for detailed execution monitoring.

2. What are the main KPIs in the Supply Chain Insight Hub?

ANSWER:

The primary KPIs are:

- TNS Achievement
- Order Fulfillment
- TAT Adherence

These KPIs are used to monitor delivery execution performance and operational efficiency.

3. How frequently is the Supply Chain Insight Hub refreshed?

ANSWER:

The Supply Chain Insight Hub is refreshed three times daily:

- 07:00
- 13:00
- 17:00

This enables near real-time operational monitoring across the delivery process.

4. What is the purpose of the Current Month Summary?

ANSWER:

The Current Month Summary monitors current month performance against customer signed targets at CBF level and assesses supply risk from unallocated orders and stock gaps.

Business Purpose:

- Identify sales risk early
- Prioritize allocation
- Accelerate picking
- Support target achievement

5. What is BTD in the Supply Chain Insight Hub?

ANSWER:

BTD stands for Balance To Do.

Logic:
BTD = MAX(Target - Billing - Picking, 0)

Business Purpose:

- Measure remaining execution gap against target
- Identify billing shortfall risk

- Support operational prioritization

6. What is the logic for Free Stock availability?

ANSWER:

Free Stock Logic:

If:
(Stock - Total Open Order) > 0

Then:
Free Stock = YES (Green)

Else:
Blank

Business Purpose:

- Identify available stock for order generation
- Support order transfer to plants with stock availability

7. Why is free stock visibility operationally important?

ANSWER:

Free stock visibility helps:

- Sales teams generate orders for available stock
- Execution teams transfer orders to plants with stock availability
- Reduce BTD risk
- Improve billing opportunity utilization

It supports faster execution and target protection.

8. What is the Delivery Loop in the Supply Chain Insight Hub?

ANSWER:

The Delivery Loop represents the operational flow from Open Orders to Customer Claims.

Flow:
Open Orders → In Delivery → Picking/Conversion → Billing → Last Mile Delivery → Customer Claims

Business Purpose:

- Monitor end-to-end order execution
- Identify operational bottlenecks
- Improve delivery visibility

9. What is the purpose of Open Orders monitoring?

ANSWER:

Open Orders monitoring tracks sales orders and allocation status against Lead Time (LT).

Business Purpose:

- Identify LT deviations
- Detect unallocated orders
- Monitor confirmation gaps
- Prevent billing impact

10. Why are LT deviations operationally risky?

ANSWER:

LT deviations indicate delays in order execution against expected timelines.

Operational Impact:

- Delayed billing

- Increased execution backlog
- Lower order fulfillment performance
- Higher delivery risk

The dashboard helps identify these risks before billing impact occurs.

11. What is monitored in the In Delivery stage?

ANSWER:

The In Delivery stage monitors:

- outbound deliveries
- credit block status
- system errors
- regional execution distribution

Business Purpose:

- Ensure confirmed deliveries are not blocked
- Improve execution control
- Identify delivery bottlenecks

12. Why are credit blocks operationally important?

ANSWER:

Credit blocks can prevent confirmed deliveries from progressing to execution.

Operational Impact:

- Delivery delays
- Billing delays
- Increased backlog
- Reduced order fulfillment efficiency

The dashboard helps identify blocked deliveries early.

13. What is the purpose of Picking / Conversion monitoring?

ANSWER:

Picking / Conversion monitoring tracks:

- Transfer Order (TO) creation
- warehouse picking progress

Business Purpose:

- Identify warehouse bottlenecks
- Monitor pending warehouse workload
- Improve delivery-to-billing conversion

14. Why is Picking / Conversion critical in the delivery process?

ANSWER:

Picking / Conversion directly impacts billing execution.

Operational Impact:

- Delayed picking slows billing
- Pending TO workload increases execution backlog
- Warehouse bottlenecks reduce delivery efficiency

The dashboard helps identify operational delays at warehouse level.

15. What is the operational meaning of Before 2 PM / After 2 PM split?

ANSWER:

The Before 2 PM / After 2 PM split measures daily warehouse operational discipline during TO execution.

Business Purpose:

- Monitor execution timeliness
- Identify delayed warehouse processing
- Improve operational responsiveness

16. What is the purpose of Billing monitoring?

ANSWER:

Billing monitoring tracks current month billing execution and regional contribution.

Business Purpose:

- Monitor billing acceleration
- Track target achievement
- Identify regional billing gaps

17. Why is MOM Billing comparison important?

ANSWER:

MOM Billing compares current month billing against previous month performance.

Business Purpose:

- Monitor billing trend
- Identify performance decline
- Evaluate billing growth or slowdown

18. What is the purpose of Last Mile Delivery monitoring?

ANSWER:

Last Mile Delivery monitoring tracks:

- delivery timeliness
- transporter performance
- shipment status

Business Purpose:

- Ensure EDD adherence
- Improve service level performance
- Monitor shipment delays

19. What is EDD?

ANSWER:

EDD stands for Expected Delivery Date.

Business Purpose:

- Measure delivery timeliness
- Monitor shipment adherence
- Evaluate transporter performance

20. Why is TAT adherence important in delivery operations?

ANSWER:

TAT adherence measures whether delivery execution is completed within expected timelines.

Operational Impact:

- Better customer service
- Improved delivery reliability
- Reduced shipment delays
- Higher operational efficiency

Poor TAT adherence increases service risk and delivery backlog.

21. What is the purpose of Customer Claims (CMS) monitoring?

ANSWER:

Customer Claims (CMS) monitoring tracks:

- claim ageing
- approval turnaround time
- departmental backlog

Business Purpose:

- Prevent claim ageing escalation
- Reduce revenue blockage
- Improve claim resolution visibility

22. Why is claim ageing operationally risky?

ANSWER:

Claim ageing increases unresolved customer exposure and delays claim closure.

Operational Impact:

- Revenue blockage
- Higher backlog
- Delayed approvals
- Reduced operational efficiency

The dashboard helps identify ageing severity and backlog concentration.

23. What is the purpose of the department-wise claim heatmap?

ANSWER:

The department-wise claim heatmap identifies functional backlog across claim processing teams.

Business Purpose:

- Identify approval bottlenecks
- Monitor claim workload concentration

- Support faster resolution

24. Is the Free Stock Indicator part of the Delivery Loop?

ANSWER:

No.

The Free Stock Indicator is not part of the Delivery Loop.

It is a separate operational visibility block used for:

- stock availability monitoring
- order generation support
- plant-wise free stock visibility

25. What are the main stages of the Supply Chain Insight Hub architecture?

ANSWER:

The architecture includes:

- Data Sources
- Data Acquisition
- Data Warehouse
- Analyze & Utilize

Key technologies:

- SAP
- Azure Storage
- Snowflake
- Python

- SQL
- Power BI

26. What is the role of Snowflake in the Supply Chain Insight Hub?

ANSWER:

Snowflake acts as the cloud data warehouse platform.

Responsibilities:

- data processing
- persistent storage
- staging
- data transformation
- pipeline orchestration

Python and SQL are used within Snowflake for transformation and automation.

27. What is the role of Power BI in the Supply Chain Insight Hub?

ANSWER:

Power BI is the primary reporting and visualization platform.

Business Purpose:

- Near real-time dashboard reporting
- Operational monitoring
- Delivery execution visibility
- KPI tracking

Power BI connects directly to Snowflake using Direct Query.

VERY IMPORTANT QUESTIONS

28. How do unallocated orders impact billing risk?

ANSWER:

Unallocated orders cannot progress into delivery and billing execution.

Operational Impact:

- Billing delays
- Increased BTD
- Lower target achievement
- Higher execution backlog

The dashboard helps identify allocation gaps early to reduce billing risk.

29. How do warehouse bottlenecks impact billing performance?

ANSWER:

Warehouse bottlenecks delay TO creation and picking completion.

Operational Impact:

- Delayed conversion from delivery to billing
- Increased pending workload
- Slower order execution
- Lower billing achievement

Picking/Conversion monitoring helps identify these bottlenecks.

30. How does free stock help reduce BTD risk?

ANSWER:

Free stock enables sales and execution teams to generate and transfer orders using immediately available inventory.

Operational Impact:

- Faster billing opportunity utilization
- Reduced execution gap
- Lower BTD
- Improved target achievement

31. Why is end-to-end delivery visibility operationally important?

ANSWER:

End-to-end delivery visibility enables monitoring across all execution stages from Open Orders to Customer Claims.

Operational Benefits:

- Faster issue identification
- Improved execution control
- Better billing visibility

- Reduced operational delays
- Improved service performance

32. Required Roles to Access DCT and Snowflake?

ANSWER:

i. IDM2BCD_SNOWFLAKE_PRXDMLB_BOSCH_DCTINDIA_DEV_GS_DEVELOPER

ii. IDM2BCD_SNOWFLAKE_PRXDMLB_BOSCH_DCTINDIA_DEV_MA_DEVELOPER